

Analyzing the retraining frequency of global forecasting models: towards more stable forecasting systems

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Outline

Forecasting Stability

Experimental Design

Empirical Results

Conclusions

Forecasting Stability

What is Forecast Stability?

Horizontal stability

- ▶ Measures how forecasts change across *different future time point* from the *same origin*.

Vertical stability

- ▶ Measures how forecasts change for the *same future time point* when the model is re-estimated at *different origins*.

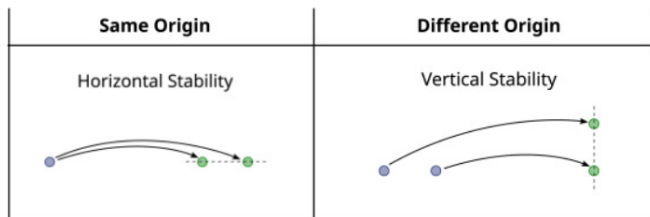


Figure 1: Godahewa, R., et al. (2025). *On forecast stability*.

Why it matters?

Forecasting drives operational decisions (inventory planning, workforce scheduling, budgeting and supply chain decisions, etc.)

Unstable forecasts lead to:

- ▶ revisions to plans and decisions,
- ▶ planning inefficiencies and higher costs,
- ▶ loss of credibility and trust in the forecasting system.

Accuracy–Stability Trade-off

Why can a trade-off arise?

- ▶ Forecasting models are updated frequently.
- ▶ More frequent retraining $\Rightarrow$ more accurate forecasts.
- ▶ Frequent model updates may produce structural model changes $\Rightarrow$ forecasts can shift.

Key questions

- ▶ Can we maintain high accuracy while producing stable forecasts?
- ▶ Is frequent retraining necessary?

Experimental Design

Datasets

3 large forecasting competition datasets: M4, M5, and VN1.

Dataset	Frequency	N. Series	T	h
M4	Daily	4.227	364	28
M5	Daily	28.298	364	28
VN1	Weekly	15.053	52	13

Forecasting Models

10 different global forecasting models, 5 classical machine learning methods and 5 deep learning architectures.

Machine Learning	Deep Learning
Linear Regression (LR)	MLP
Random Forest (RF)	LSTM
XGBoost	TCN
LGBM	NBEATS
CatBoost	NHITS

Evaluation Strategy: Rolling Origin

- ▶ **Why?** Out-of-sample testing assesses generalization
- ▶ **How?**
 - ▶ Train/test split, preserving temporal order
 - ▶ Train model on initial training set
 - ▶ Forecast h steps ahead
 - ▶ Shift origin forward by p steps
 - ▶ Repeat until the test set is exhausted
- ▶ **Window types:**
 - ▶ Fixed window: Only the most recent n observations used
 - ▶ Expanding window: Include all historical data up to the current origin
- ▶ **In our study:** Expanding window with a step size $p = 1$

Evaluation Metrics

Accuracy

- ▶ **Point:** Root Mean Squared Scaled Error (RMSSE) (1)
- ▶ **Probabilistic:** Scaled Multi-Quantile Loss (SMQL) (2),(3)

Stability

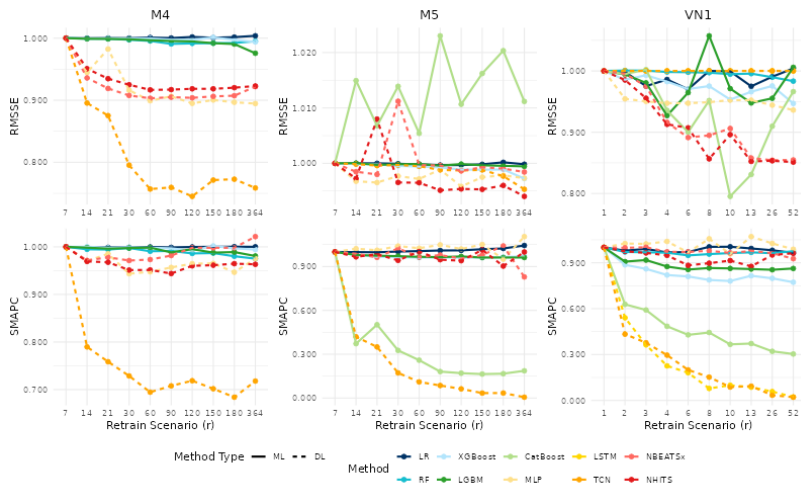
- ▶ **Point:** Simmetric Mean Absolute Percentage Change (sMAPC) (4)
- ▶ **Probabilistic:** Scaled Multi-Quantile Change (SMQC) (5),(6)

Retraining Scenarios

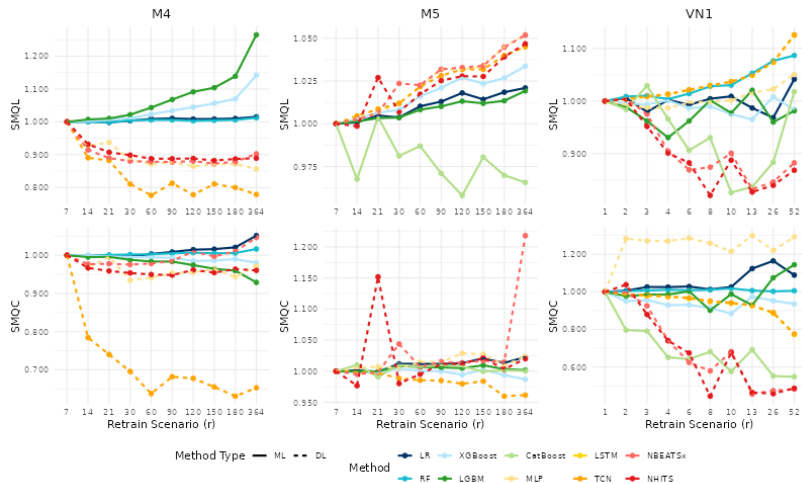
- ▶ **Retrain window (r):** the interval at which the model is re-trained or updated.
- ▶ **Scenarios based on data frequency:**
 - ▶ Daily: $r = \{7, 14, 21, 30, 60, 90, 120, 150, 180, 364\}$
 - ▶ Weekly: $r = \{1, 2, 3, 4, 6, 8, 10, 13, 26, 52\}$
- ▶ **Three strategies:**
 - ▶ Continuous retraining, $r = 7$ or $r = 1$, baseline scenario
 - ▶ Periodic retraining, $7|1 < r < T$
 - ▶ No retraining, $r = T$

Empirical Results

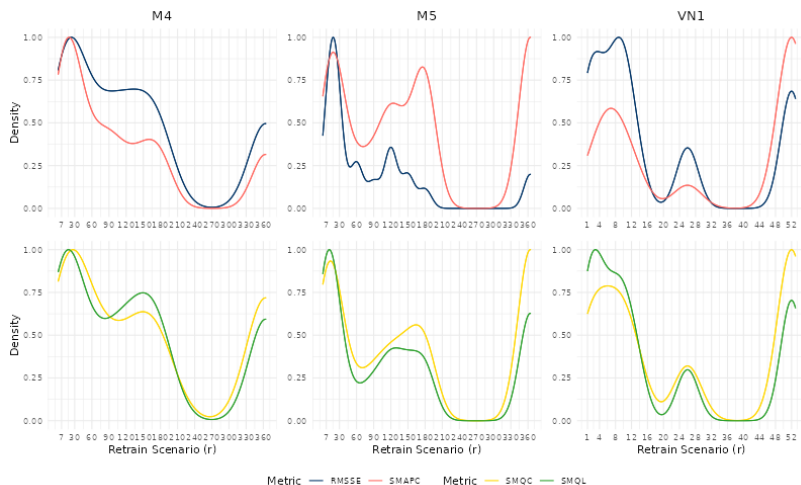
Point Forecasting



Probabilistic Forecasting



Optimal Retraining Frequency



Conclusions

Key Results

Accuracy

- ▶ Point accuracy is stable
- ▶ Probabilistic accuracy slightly degrades with less frequent retraining

Stability

- ▶ Point stability shows improvements for lower retraining frequencies
- ▶ Probabilistic stability on average is stable

Optimal Retraining

- ▶ Some form of periodic retraining is always better than continuous retraining

Key Takeaways

- ▶ Frequent retraining is not necessary (under stable demand)
- ▶ Periodic retraining strategies allow for balancing accuracy and stability
- ▶ Reducing the retraining frequency supports more sustainable forecasting systems (lower costs of producing forecasts).

References

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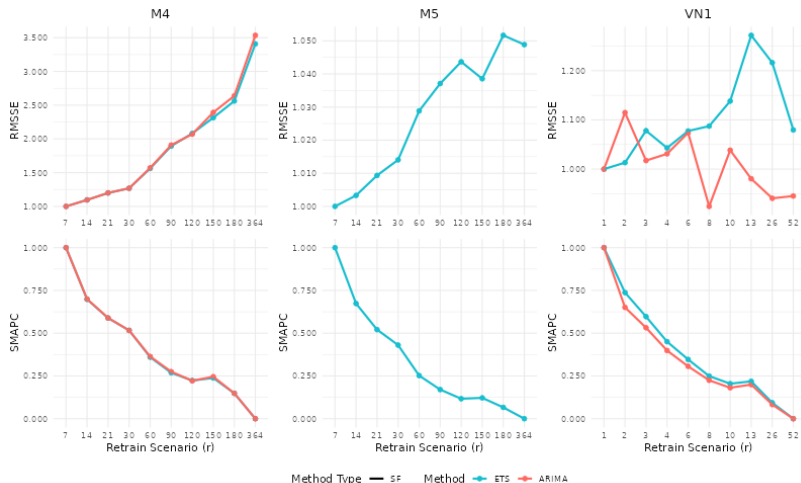
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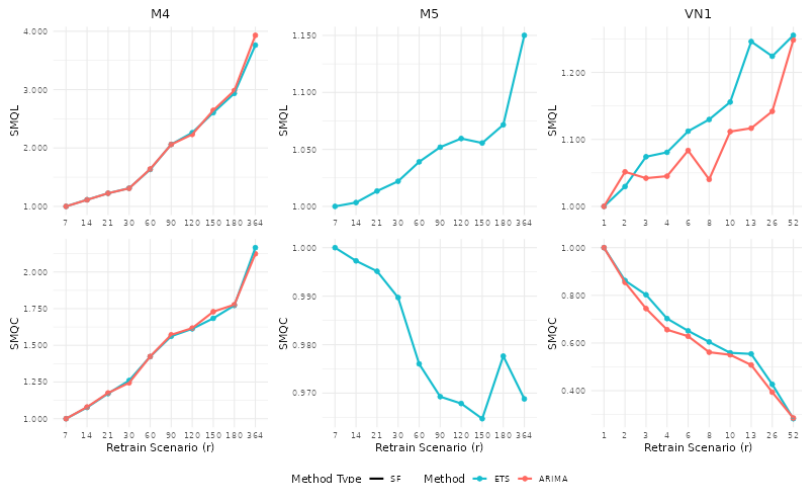
Thank you!

Appendix

Local Models - Point Forecasting



Local Models - Probabilistic Forecasting



Accuracy Metrics - Math

$$\text{RMSSE} = \sqrt{\frac{\frac{1}{h} \sum_{t=n+1}^{n+h} (y_t - \hat{y}_t)^2}{\frac{1}{n-s} \sum_{t=s+1}^n (y_t - y_{t-s})^2}} \quad (1)$$

$$\text{SQL}(\alpha) = \frac{\frac{1}{h} \sum_{t=n+1}^{n+h} \left(\alpha \cdot (y_t - \hat{q}_t^{(\alpha)}) \cdot \mathbb{I}_{\hat{q}_t^{(\alpha)} \leq y_t} + (1-\alpha) \cdot (\hat{q}_t^{(\alpha)} - y_t) \cdot \mathbb{I}_{\hat{q}_t^{(\alpha)} > y_t} \right)}{\frac{1}{n-s} \sum_{t=s+1}^n |y_t - y_{t-s}|} \quad (2)$$

$$\text{SMQL} = \frac{1}{|Q|} \sum_{\alpha \in Q} \text{SQL}(\alpha) \quad (3)$$

Stability Metrics - Math

$$\text{sMAPC} = \frac{200}{h-1} \sum_{t=n+1}^{n+h-1} \frac{|\hat{y}_{t,n} - \hat{y}_{t,n-1}|}{|\hat{y}_{t,n}| - |\hat{y}_{t,n-1}|} \quad (4)$$

$$\text{SQC}(\alpha) = \frac{\frac{1}{h-1} \sum_{t=n+1}^{n+h-1} \left(\alpha \cdot (\hat{q}_1 - \hat{q}_2) \cdot \mathbb{I}_{\hat{q}_2 \leq \hat{q}_1} + (1-\alpha) \cdot (\hat{q}_2 - \hat{q}_1) \cdot \mathbb{I}_{\hat{q}_2 > \hat{q}_1} \right)}{\frac{1}{n-s} \sum_{t=s+1}^n |y_t - y_{t-s}|} \quad (5)$$

where $\hat{q}_1 = \hat{q}_{t,n-1}^{(\alpha)}$; $\hat{q}_2 = \hat{q}_{t,n}^{(\alpha)}$

$$\text{SMQC} = \frac{1}{|\mathcal{Q}|} \sum_{\alpha \in \mathcal{Q}} \text{SQC}(\alpha) \quad (6)$$

Computing Setup

- ▶ Microsoft Azure NC6s v3 VM
- ▶ 6 vCPUs, 1 GPU, 112 GB RAM
- ▶ Nixtla's libraries: `statsforecast`, `mlforecast`, `neuralforecast`